

Total: /20

- 1) Would $f(x) = \frac{x-3}{x^2+3}$ have any vertical/horizontal asymptotes? Justify your response with use of limits.

There is no number that can make the denominator 0, so no vertical asymptote.
For horizontal asymptotes
 $\lim_{x \rightarrow \pm\infty} \frac{x(1-\frac{3}{x})}{x^2(1+\frac{3}{x^2})} = \frac{1}{x} \lim_{x \rightarrow \pm\infty} \frac{1-\frac{3}{x}}{1+\frac{3}{x^2}} = \frac{1}{\infty} = 0^+$
 $\lim_{x \rightarrow -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0^-$ (2)

$\therefore$ Horizontal asymptote of $y = 0$

- 2) What are all the possible graphical interpretations of a critical point that is not a horizontal tangent? Provide illustrations as part of your explanation.

CP = A point where $f'(x) = 0$ or undefined and $f(x)$ exists

These 2 have a H.T., so they don't count for this question

smooth local max
smooth local min

corner $\rightarrow$ not counted for this question
 $\rightarrow$ different slope for each side, $\therefore$ und. (2) 1.5

vertical tangent $\rightarrow$ vertical tangent has no slope, $\therefore$ und.

cusp
V.A.?

marked wrong, no VA

(The dots are the critical points)

- 3) Use the first derivative test to classify any local extrema for $y = (x-3)^3 - 12x$.

$$\begin{aligned} \frac{dy}{dx} &= 3(x-3)^2 - 12 \\ &= 3x^2 - 18x + 27 - 12 \\ &= 3x^2 - 18x + 15 \\ &= 3(x^2 - 6x + 5) \\ &= 3(x-1)(x-5) \rightarrow x=1, 5 \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= 3(1-1)(1-5) = 0 \\ \text{when } x=1, y &= (1-3)^3 - 12(1) = -8 - 12 = -20 \end{aligned}$$

$$\begin{aligned} \text{when } x=5, y &= (5-3)^3 - 12(5) = 8 - 60 = -52 \end{aligned}$$

	$(-\infty, 1)$	$(1, 5)$	$(5, \infty)$
f'	+	+	+
f	inc	dec	inc

(1, y) smooth local max
(5, y) smooth local min
(4) 3.5

has to be smooth because polynomial

$\therefore$ local extrema: $(1, 20)(5, -52)$

- 4) Given the function $f(x) = \frac{x^3 - 1}{x^2 + 2x}$, using methods presented in Calculus,
- a) determine all applicable asymptotes and the corresponding behavior around the asymptotes. (For V.A check behavior on either side, for H.A or O.A check if function lies above or below). You do not need to do first derivative test - just use test point)

$$f(x) = \frac{(x-1)(x^2+x+1)}{x(x+2)} = \frac{(x-1)(x+1)^2}{x(x+2)} \xrightarrow{x \neq -1} x = 0, -2$$

$$\lim_{x \rightarrow 0^-} f(x) = \infty \quad \lim_{x \rightarrow -2^-} f(x) = -\infty$$

$$\lim_{x \rightarrow 0^+} f(x) = -\infty \quad \lim_{x \rightarrow -2^+} f(x) = \infty$$

Oblique

$$\begin{array}{r} x^2 + 2x + 0 \overline{) x^3 + 0x^2 + 0x - 1} \\ \underline{x^3 + 2x^2} \\ -2x^2 + 0x - 1 \\ \underline{-2x^2 - 4x + 0} \\ 4x - 1 \end{array}$$

$$\left(\frac{1}{4} - 2\right) = -\frac{7}{4}$$

$4x - 1 \rightarrow$ crosses at $\left(\frac{1}{4}, -\frac{7}{4}\right)$ or $(0.25, -1.75)$

HA behaviour on oblique, $(100, 98)$ $(-100, -102)$
 in function $f(100) = \frac{999999}{10200} = 98.01$
 $f(-100) = \frac{-1000001}{10200} = -102.04$

$$\lim_{x \rightarrow \infty} f(x) = 98^+ \quad \lim_{x \rightarrow -\infty} f(x) = -102^-$$

above below

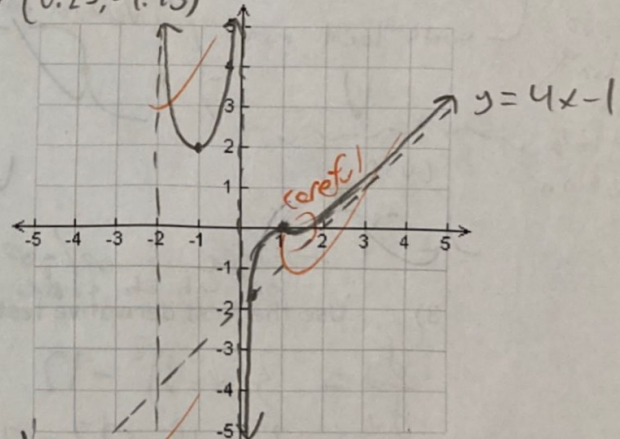
- b) sketch the graph.

assumed critical points $x = -2.5, 1$

$$f(-2.5) = \frac{-16.625}{1.25} = -13.3 \quad (-2.5, -13.3)$$

$$f(-1) = \frac{-2}{-1} = 2 \quad (-1, 2)$$

$$f(1) = 0 \quad (1, 0)$$



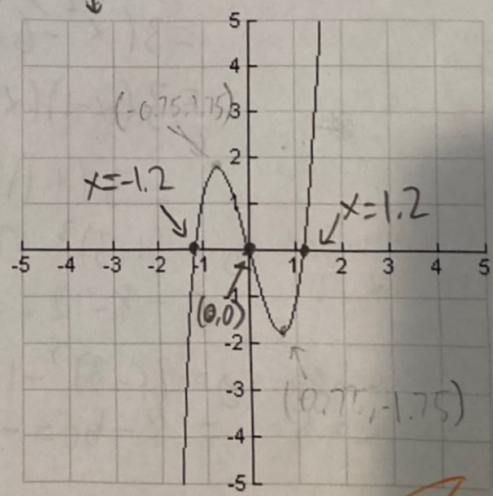
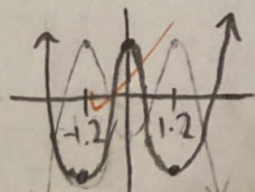
- 5) The function below represents the derivative function $f'(x)$ of a function $f(x)$. State

- i) intervals where $f(x)$ is increasing:
 $x \in (-1.2, 0) \cup (1.2, \infty)$

- ii) intervals where $f(x)$ is decreasing:
 $x \in (-\infty, -1.2) \cup (0, 1.2)$

- iii) x-coordinate(s) of local extrema:
 $x = -1.2, 0, 1.2$

- b) Make a rough sketch of the corresponding $f(x)$.



- 6) Determine the values of a , b , c , and d , given $g(x) = ax^3 + bx^2 + cx + d$ has a local minimum at $(-1, 1)$ and a local maximum at $(0, 5)$.

$$g'(-1) = 0 \quad g'(0) = 0 \quad g(0) = 5$$

$$g'(x) = 3ax^2 + 2bx + c$$

$$5 = 0 + 0 + 0 + d$$

$$\underline{d = 5}$$

$$g'(-1) \rightarrow 0 = 3a - 2b + c \rightarrow 3a - 2b = 0$$

$$g'(0) \rightarrow 0 = 0 + 0 + c \rightarrow \underline{c = 0}$$

$$g(-1) \rightarrow 1 = -a + b + 5$$

$$b = -4 + a \rightarrow b = -4 - 8 = -12$$

$$\underline{b = -12}$$

$$3a - 2b = 0$$

$$3a - 2(-4 + a) = 0$$

$$3a + 8 - 2a = 0$$

$$\underline{a = -8}$$

$$\therefore a = -8 \quad b = -12 \quad c = 0 \quad d = 5$$

(3)